Gas exchange in humans

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1

Core | Gas exchange in humans | [1]

Which list gives only features of a gas exchange surface in humans?

A. thick surface, large surface area, good blood supply

B. thin surface, large surface area, good ventilation with air

C. small surface area, thin surface, good blood supply

D. thin surface, poor blood supply, good ventilation with air

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2 · Part 1

Core | Gas exchange in humans | [1]

The blood in the capillaries around the alveoli is constantly carried away and replaced by more blood.

How does this good blood supply help gas exchange?

A. It makes the wall of each alveolus thinner, so oxygen diffuses a shorter distance.

B. It increases the surface area of the alveoli, so more oxygen diffuses at the same time.

C. It pumps oxygen from the air in the alveoli into the blood.

D. It keeps the oxygen concentration in the blood lower than in the air in the alveoli, so oxygen keeps diffusing into the blood.

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3 · Part 1 Core Gas exchange in humans [1]

Which row shows how expired air differs from inspired air?

	oxygen	carbon dioxide	water vapour
A	less	more	more
B	less	more	less
C	more	less	more
D	more	less	less

4 · Part 1 Core Gas exchange in humans [1]

Two identical test-tubes, 1 and 2, each contain 10 cm^3 of limewater.

Priya breathes out gently through a straw into the limewater in tube 1. Using a hand pump, she bubbles the same volume of air from the room through the limewater in tube 2.

Which result and conclusion are correct?

A Tube 2 turns milky first, which shows that inspired air contains more carbon dioxide than expired air.

B Both tubes turn milky at the same time, which shows that breathing does not change the amount of carbon dioxide in air.

C Tube 1 turns milky first, which shows that expired air contains more carbon dioxide than inspired air.

D Tube 1 turns milky first, which shows that expired air contains less oxygen than inspired air.

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5 · Part 1 Core Gas exchange in humans [1]

Tomas measures his breathing at rest and again straight after running for five minutes. The volume of air per breath is a measure of the depth of breathing.

activity	breathing rate / breaths per minute	volume of air per breath / dm^3
at rest	14	0.5
after running	30	1.8

Which statement describes the effect of running on Tomas's breathing?

A Both the rate and the depth of breathing increased.

B The rate of breathing increased but the depth of breathing stayed the same.

C The depth of breathing increased but the rate of breathing decreased.

D The rate of breathing increased but the depth of breathing decreased.

6 · Part 1 Extended Gas exchange in humans [1]

What is the function of the cartilage in the wall of the trachea?

A to trap dust and pathogens before they reach the lungs

B to move mucus up towards the throat

C to allow gases to diffuse between the air and the blood

D to keep the trachea open so that air can flow freely

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7 · Part 1 Extended Gas exchange in humans [1]

Which row describes what happens when a person breathes in?

	external intercostal muscles	diaphragm	volume of thorax	pressure in thorax
A	relax	relaxes	decreases	increases
B	contract	contracts	increases	decreases
C	contract	relaxes	increases	increases
D	relax	contracts	decreases	decreases

8 · Part 1 Extended Gas exchange in humans [1]

Which statement correctly explains a difference between inspired air and expired air?

A Expired air contains less oxygen because oxygen diffuses from the blood into the air in the alveoli.

B Expired air contains less oxygen because oxygen molecules are changed into carbon dioxide molecules in the air in the alveoli.

C Expired air contains more water vapour because water evaporates from the moist lining of the alveoli.

D Expired air contains more carbon dioxide because carbon dioxide diffuses from the air in the alveoli into the blood.

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9 · Part 1 Extended Gas exchange in humans [1]

During physical activity, the rate and depth of breathing increase.

What causes this change?

A The brain detects a decrease in the concentration of carbon dioxide in the blood.

B The brain detects an increase in the concentration of carbon dioxide in the blood.

C The lungs detect an increase in the concentration of oxygen in the air in the alveoli.

D The leg muscles detect an increase in the concentration of carbon dioxide and send a message to the diaphragm.

10 · Part 1 Extended Gas exchange in humans [1]

Which row shows the roles of goblet cells and ciliated cells in protecting the breathing system?

	goblet cells	ciliated cells
A	make mucus that traps pathogens and particles	move mucus up towards the throat
B	move mucus up towards the throat	make mucus that traps pathogens and particles
C	make mucus that traps pathogens and particles	engulf and digest the trapped pathogens
D	engulf and digest dust particles	move mucus down into the alveoli

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11 · Part 1 Core Gas exchange in humans [5]

The lungs of an adult contain about 480 million alveoli. Together they give a gas exchange surface with an area of about 70 m^2.

(a) State two features of a gas exchange surface in humans, other than a large surface area. [2]

(b) Explain why a large surface area is important for gas exchange. [1]

(c) In one lung disease, the thin walls between neighbouring alveoli break down. Many small alveoli join together to form fewer, larger air spaces.

Suggest and explain why a person with this disease becomes out of breath when walking. [2]

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12 · Part 1 Core Gas exchange in humans [6]

The table shows the percentages of two gases in inspired air and in expired air.

gas	inspired air / %	expired air / %
oxygen	21	16
carbon dioxide	0.04	4

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(a) Using data from the table, describe how the composition of expired air differs from the composition of inspired air. [2]

(b) Calculate how many times greater the percentage of carbon dioxide is in expired air than in inspired air. [2]

(c) State how the amount of water vapour in expired air compares with the amount in inspired air. [1]

(d) On a cold day, a cloud of tiny water droplets can be seen when a person breathes out.

Suggest why. [1]

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13 · Part 1 Core Gas exchange in humans [6]

Leila sets up two boiling tubes, P and Q, each containing 15 cm^3 of limewater. Both tubes are joined to one mouthpiece by glass and rubber tubing. When Leila breathes in through the mouthpiece, air from the room is drawn through the limewater in tube P. When she breathes out through the mouthpiece, her breath is forced through the limewater in tube Q.

Leila counts the number of breaths until the limewater in each tube turns milky.

tube	number of breaths until the limewater turned milky
P	not milky after 40 breaths
Q	6

(a) Name the gas that turns limewater milky. [1]

(b) Explain what the results show about inspired air and expired air. [2]

(c) State one variable, other than the volume of limewater, that should be kept the same for both tubes. [1]

(d) Suggest why this apparatus cannot be used to compare the oxygen content of inspired air and expired air. [1]

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(e) Suggest why the limewater in tube P would turn milky if Leila carried on breathing through the apparatus for much longer. [1]

14 · Part 1 Core Gas exchange in humans [4]

Sam counts the number of breaths he takes in 30 seconds while sitting still. He then does step-ups onto a bench for two minutes and immediately counts his breaths for another 30 seconds.

activity	number of breaths in 30 s
sitting still	8
straight after step-ups	19

(a) Calculate Sam's breathing rate straight after the step-ups, in breaths per minute. [1]

(b) Describe the effect of the step-ups on Sam's rate of breathing, and state the effect on his depth of breathing. [2]

(c) Suggest how Sam could make his results more reliable. [1]

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15 · Part 1 Core Gas exchange in humans [6]

The table shows measurements of a woman's breathing during three activities.

The volume of air breathed in per minute is found by multiplying the breathing rate by the volume of air per breath.

activity	breathing rate / breaths per minute	volume of air per breath / dm^3
sitting	12	0.50
walking	18	1.20
running	30	2.40

(a) Calculate the volume of air the woman breathes in per minute while running. Give the unit. [2]

(b) Calculate the percentage increase in the volume of air breathed in per minute when the woman changes from sitting to walking. [2]

(c) Her friend says: "The extra air breathed in per minute when running, compared with sitting, is caused only by the faster breathing rate."

Use the data to explain why her friend is wrong. [2]

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16 · Part 1 Extended Gas exchange in humans [6]

Breathing in and breathing out are caused by muscles that change the volume of the thorax.

(a) Explain how the diaphragm and the external intercostal muscles cause air to enter the lungs. [4]

(b) During forced breathing out, for example when coughing, the internal intercostal muscles contract.

State how this moves the ribs. [1]

(c) State what happens to the diaphragm when a person breathes out. [1]

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17 · Part 1 Extended Gas exchange in humans [5]

The lungs are inside the thorax, which is normally an airtight space.

In an accident, a hole is made through the chest wall on the left side. Air from outside can now move freely into the space around the left lung.

(a) State what happens to the volume and to the pressure inside the thorax when a healthy person breathes in. [2]

(b) The person tries to breathe in, but the left lung does not fill with air.

Suggest why. [3]

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18 · Part 1 Extended Gas exchange in humans [6]

The table shows the percentages of oxygen and carbon dioxide in inspired air, in the air inside the alveoli and in expired air.

gas	inspired air / %	air in alveoli / %	expired air / %
oxygen	21	14	16
carbon dioxide	0.04	5	4

(a) Explain why the air in the alveoli contains less oxygen than inspired air. [2]

(b) Explain why the air in the alveoli contains more carbon dioxide than inspired air. [2]

(c) No gas exchange happens in the trachea and bronchi. When a person breathes out, the air leaving the alveoli mixes with the air in the trachea and bronchi.

Suggest why expired air contains more oxygen and less carbon dioxide than the air in the alveoli. [2]

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19 · Part 1 Extended Gas exchange in humans [5]

During a cycle race, the rate and depth of a cyclist's breathing increase.

(a) Explain why the concentration of carbon dioxide in the cyclist's blood increases during the race. [2]

(b) Name the organ that detects the increase in the concentration of carbon dioxide in the blood. [1]

(c) In a laboratory, a person sitting still breathes in air containing 5% carbon dioxide for a short time.

Predict and explain the effect on the person's breathing. [2]

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20 · Part 1 Extended Gas exchange in humans [6]

The trachea and bronchi are lined with goblet cells and ciliated cells.

(a) Describe the role of goblet cells in protecting the breathing system. [2]

(b) Explain how ciliated cells help to protect the breathing system. [2]

(c) In a rare inherited condition, the cilia on the ciliated cells cannot move. People with this condition have many more lung infections than other people.

Suggest why. [2]

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